ZIMBABWE INTEGRATING CAPITAL AND RISK PROGRAMME

('ZICARP')

TS 6.1 – Determination of Market Risk Capital Requirement

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1. Introduction

1.1 Market risk is the risk of loss to an insurer's assets and liabilities arising from the level or volatility of market prices of financial instruments.

1.2 The impact of changes in financial variables such as stock prices, interest rates, property prices and currency rates is used to gauge exposure to market risk.

1.3 However, Zimbabwe does not have an active derivative market. Therefore, liquidity risk and single equivalent scenario adjustment for management actions are not applicable.

1.4 Under the Zimbabwe Integrating Capital and Risk Programme ('ZICARP'), we classify market risk into five sub risk modules:

i. equity price risk;
ii. property risk;
iii. counterparty/default risk;
iv. currency risk; and
v. concentration risk.

1.5 The objectives of the Solvency Capital Requirement ('SCR') for market risk are based on specified stress scenarios applied to the above market risk modules .

1.6 Diversification within the above risk modules is further allowed for in the determination of market risk.

1.7 Equity risk derives from the volatility of the market prices of equities.

1.8 Property risk arises from volatility in the market prices of property.

1.9 The counterparty/default risk under ZICARP reflects all possible losses that may arise due to unexpected default or deterioration in credit worthiness of an insurer's counterparties and debtors over the next twelve months.

1.10 Currency risk derives from movements in the level and volatility of currency exchange rates and their impact on the insurer's assets and liabilities.

1.11 Concentration risk is the potential of a loss due to movements in equity, spread and default risk and property risk on assets that are considered to be not sufficiently diversified and where there is over exposure.

2. Framework for Market Risk

2.1 The overall market risk capital requirement is calculated by the formula:

$$SCR_{Mkt} = \sqrt{\sum_i \sum_j CorrMkt_{i,j} * Mkt_i * Mkt_j}$$

where:

i. Mkt_i = the market risk sub-module i

ii. Mkt_j = the market risk sub-module j iii.

iii. $CorrMkt_{i,j}$ = the correlation parameter between components i and j (refer to section 2.6, Table 1 below).

2.2 In the actual calculation, Mkt_i and Mkt_j will be replaced by:

$\mathrm{Mkt}_{\mathrm{eqprice}}$ which denotes the equity risk sub-module;

Mkt_{prop} which denotes the property risk sub-module;

$SCR_{default}$ which denotes the counterparty/default risk sub-module;

Mkt_{fx} which denotes the currency risk sub-module; and

$\mathrm{Mkt}_{\mathrm{conc}}$ which denotes the concentration risk sub-module.

2.3 The formula for Market Risk is adopted from the global Solvency II regime and simplifications for calculating sub-risk modules are adopted from the Solvency Assessment and Management ('SAM') regime in South Africa.

2.4 Parameters in the formula will continue to evolve through application of actuarial control cycle principles during implementation of ZICARP.

2.5 The correlation matrix for market risk includes each of the market sub risk modules, excluding the illiquidity premium risk.

2.6 The proposed correlation matrix for Zimbabwe is presented in the Table below.

	Equity	**Property**	**Counterparty/Default**	**Currency**	**Concentration**
Equity	100%	75%	75%	50%	50%
Property		100%	50%	50%	50%
Counterparty/ Default			100%	50%	50%
Currency				100%	50%
Concentration					100%

Table 1: Correlation matrix for market risk

2.7 An additional 20% capital charge will be added for those investments beyond investments limits. Investments limits will be prescribed from time to time. The asset values used under market risk must be those after investment limits have been applied.

3. Calculation of Capital Requirements for the Sub-Risk Modules

Equity Price Risk

3.1 Equity price risk arises when the market values of assets and liabilities are sensitive to changes in the market prices for equities. The capital requirement for equity risk ($\mathrm{Mkt_{eqprice}}$) is calculated as:

$$\mathrm{Mkt_{eqprice}} = \sqrt{\sum_{i,j} \mathrm{CorrEq_{i,j}} * \mathrm{Mkt_{eq,i}} * \mathrm{Mkt_{eq,j}}}$$

where:

$\mathrm{CorrEq_{i,j}}$ = The entries of the correlation matrix CorrEq

Correlation matrix for equities (CorrEq)

	Zimbabwean	Foreign
Zimbabwean	1	0.75
Foreign	0.75	1

$\mathrm{Mkt_{eq,i}}$= Capital requirement for the individual market risk component *i* according to the rows of correlation matrix $\mathrm{CorrMkt}$ and

$\mathrm{Mkt_{eq,j}}$= Capital requirement for the individual market risk component *j* according to the columns of correlation matrix $\mathrm{CorrMkt}$.

Categories of Equities

3.2 Equities must be split into one of the following two categories when determining the capital requirement for equity price risk:

3.2.1 Zimbabwean equities – this covers two subcategories of

i. Quoted equities i.e., equities traded in local currency – covering equities listed on the Zimbabwe Stock Exchange and Victoria Falls Stock Exchange (VFX)

ii. Unquoted equity – these are equities that are no longer traded on a stock exchange or might be unquoted because their firm are too small to qualify for a stock market listing, have too few shareholders for a listing, or have been delisted.

3.2.2 Foreign global equities – covering equities listed in regulated markets in foreign countries.

3.3 Dual-listed shares should be included in the category that represents the exchange on which the shares were bought.

3.4 The capital requirement for equity price risk in relation to each equity component *i* must be calculated as:

$$Mkt_{eq,i} = max(\Delta BOF | Equity\ shock\ i, 0)$$

where;

i. ΔBOF= Change in Basic Own Funds;

ii. $Equity\ shock\ i$= A prescribed fall in the value of equities in category *i*.

iii. The prescribed fall in the value of equities in category *i* ($equity\ shockprice\_i$) under the $equity\ shock\ i$ stress scenario must be calculated as:

$$equity\ shockprice\_i = base\ equity\ shock\ i\ +\ symmetric\ adjustment\ i$$

Where the $base\ equity\ shock\ i$ parameters for each category *i* are set out in the table below:

Equity Category		**Shock**
Zimbabwean Equities	Quoted	40% + symmetric adjustment factor local
	Unquoted	45% + symmetric adjustment factor other
Foreign Global equities		30%

Table 2: Shocks for equities

Symmetric adjustment

3.5 The symmetric adjustment factor for each category i, except for Infrastructure assets ($symmetric\ adjustment\ i$), must be calculated as:

$$symmetric\ adjustment\ i = min\left[10\%, max\left(-10\%, 50\%.\left(\frac{CI_i - AI_i}{AI_i} - b_i\right)\right)\right]$$

where:

CI_i = Current index value for category i, where the index for category i is set out in the table below;

AI_i = The three-year moving monthly average index (equal weightings) for category i, where the index for category i is set out in the table below; and

b_i = The parameters are set out in the table below:

Equity Category	Index for category *i*	b_i
Zimbabwean Equities	Quoted	8%
	Unquoted	8%
Foreign Global Equities		0%

Table 3: Parameter for symmetric adjustment

3.6 Guidance on the calculation of symmetric adjustment factors for the different equity categories is given under appendix TS 6.1 A2 of this technical specification.

Property risk

3.7 Property risk arises when the market values of assets and liabilities are sensitive to changes in the level of market prices of property. The most significant part of this risk is the risk of economic loss due to the effect of the volatility and uncertainty of future interest rates on the mismatch of cash flows from interest sensitive assets with liability cash flows.

3.8 All assets and liabilities exposed to property risk should be included in the calculation of the property risk capital requirement, including:

3.8.1 Land, buildings and immovable-property rights; and

3.8.2 Property investment for the own use of the insurer.

3.9 For clarity, property-related investments[1] must be considered under equity risk capital charges.

3.10 The capital requirement for property risk (Mkt_{prop}) is calculated as:

$$Mktprop = \ max(\Delta BOF \,|\, property\ shock, 0)$$

Where:

i. ΔBOF = the change in the value of Basic Own Funds

ii. Property shock = an instantaneous decrease of 25% in property values.

3.11 The stress should be replaced by an equal but opposite shock if the application of the property shock stress scenario results in a capital need for property risk that is less than zero (i.e., if $Mkt_{prop} < 0$). Any short positions in property must be taken into account in this situation, even if they don't meet the criteria for an appropriate risk mitigation strategy.

3.12 Where a causal relationship exists between property price changes and policyholder behaviour, the calculation of the property risk capital requirement may take into account changes to policyholder behaviour.

[1] Property related investments can be treated as consisting of the following:

i. An investment in a company engaged in real estate management, real estate development or similar activities;

ii. An investment in a company which borrowed funds external to the insurance group to leverage investments in properties; and

iii. Direct or indirect participations in real estate companies that generate periodic income or which are otherwise intended for investment purposes.

Counterparty/default Risk

3.13 The counterparty default risk under ZICARP reflects all possible losses that may arise due to unexpected default or deterioration in credit worthiness of an insurer's counterparties and debtors over the next twelve months.

3.14 The counterparty default risk of an insurer arises from the following sources:

3.14.1 Reinsurance and similar risk-mitigating contracts such as securitisation;

3.14.2 Receivables from intermediaries; and

3.14.3 Credit exposures arising from the following:

i. Policyholder debtors,

ii. Cash at bank,

iii. Capital, initial funds, letters of credit (and any other called up but unpaid commitments), and

iv. Guarantees, letters of credit etc that are provided by the undertaking as well as any other commitments which the undertaking has provided, and which depend on the credit standing of a counterparty.

3.15 Counterparty exposures must be divided into two categories:

3.15.1 Category 1 exposures – which relates to exposures arising from risk mitigating contracts with counterparties that are likely to have credit rating. This includes reinsurance arrangements, deposits at banks and securitisation arrangements. Exposures under this category are assessed based on the probability of default and loss given default related to the credit rating of the counterparty.

3.15.2 Category 2 exposures – relates to all the other credit exposures not classified as Category 1. This includes receivables from intermediaries and policyholders' debtors.

3.16 The overall solvency capital requirement for counterparty risk ($SCR_{default}$) is calculated by the formula:

$$SCR_{default} = \sqrt{\sum_i \sum_j CorrDefault_{i,j} * SCR_{defaulty,i} * SCR_{default,j}}$$

Where:

i. $SCR_{default,i}$ = the total exposure from counterparties in Category i;

ii. $SCR_{default,j}$ = the total exposure from counterparties in Category j; and

iii. $Corr_{i,j}$ = the correlation parameter between categories i and j.

3.17 $Corr_{i,j}$ is given by the following matrix:

	$SCR_{default,1}$	$SCR_{default,2}$
$SCR_{default,1}$	1	0.75
$SCR_{default,2}$	0.75	1

3.18 Counterparty /Default risk arises from potential losses due to credit default events, such as the default of the counter party or issuer of a financial instrument held by an insurer. Default risk under ZICARP is divided into Category 1 and Category 2 exposures.

Counterparty/ default risk – Category 1 exposures

3.19 The capital requirement for default risk from Category 1 exposures is calculated by the formula:

$$SCR_{defaulty,1} = \sum_{all\ i} p_i * LGD_i * EAD_i$$

Where:

LGD_i = Loss Given Default for counterparty i;

p_i = The probability of default assigned to counterparty i; and

EAD_i = Exposure at Default of counterparty i. This is defined as the expected recovery from counterparty i less any market value of collateral, if any, in relation to the expected recovery from counterparty i.

3.20 Guidance on probability of default and loss given default for rated Category 1 counterparties is given in appendix **TS 6.1 A1.**

Counterparty default risk – Category 2 exposures

3.21 The capital requirement for default risk from Category 2 exposures is calculated by the formula:

$$SCR_{defaulty,2} = \Delta BOF|15\%.\, E + 90\%.\, E_{past\ due}$$

Where:

i. $E_{past\ due}$ = sum of all Category 2 exposures that are more than 90 days old and

ii. E = sum of all the other Category 2 exposures than are less than 90 days;

Currency risk

3.22 Currency risk arises when the market values of assets and liabilities are sensitive to changes in currency exchange rates. The capital requirement for currency risk (Mkt_{fx}) is calculated as follows:

$$Mkt_{fx} = max(Mktup_{fx}, Mktdown_{fx}, 0)$$

Where:

i. $Mktup_{fx} = max(\Delta BOF|fxupwardshock, 0)$;

ii. $Mktdown_{fx} = max(\Delta BOF|fxdownwardshock, 0)$;

iii. $fxupwardshock$ = An instantaneous increase of 50% in the value of all currencies against the official currency. The post-shock exchange rate under this scenario (Eup) should be calculated as:

iv. Eup=E/ (1+0.5), where E is the relevant exchange rate prior to the shock;

v. $fxdownwardshock$ =An instantaneous decrease of 30% in the value of all

vi. Currencies against the official currency. The post-shock exchange rate under this scenario ($Edown$) should be calculated as: $Edown$=E/ (1−0.3); and

vii. ΔBOF = The change in the value of Basic Own Funds.

3.23 In calculating the capital requirement for currency risk, an insurer may assume that management actions can take place under the stress scenarios, including changes to future bonus rates on policies with discretionary participation features.

3.24 The type and extent of management actions that an insurer may assume in the stresses must consider whether the stress is assumed to be insurer-specific or industry-wide. For currency risk, insurers should assume that the stresses result entirely from industry-wide events.

3.25 Where a causal relationship exists between exchange rate movements and policyholder behaviour, the calculation of the currency risk capital requirement may take into account changes to policyholder behaviour.

3.26 All assets and liabilities that are exposed to movements in exchange rates should be included in the calculation of the currency risk capital requirement. This should include any investment in instruments denominated in a foreign currency. For clarity:

i. Unlisted equity and property located outside of Zimbabwe should be included in the calculation of the currency risk capital requirement;

ii. Dual-listed shares which are listed on any Zimbabwe Stock Exchange and purchased on an offshore exchange should be included in the calculation of the currency risk capital requirement; and

iii. All other dual-listed shares which are listed on Zimbabwe Stock Exchange may be excluded from the calculation of the currency risk capital requirement.

Concentration risk

3.27 Concentration risk refers to the risk of potential losses on investments over and above the systematic risks arising from the portfolio of investments when the portfolio of investments is not sufficiently diversified. The capital requirement for concentration risk (Mkt_{conc}) is calculated as:

$$Mkt_{conc} = \sqrt{\sum_i Conc_i^{\,2}}$$

Where:

i. $Conc_i = \Delta BOF \mid concentration\ shock\ i$

ii. $Concentration\ shock\ i$ = an instantaneous decrease in the value of E_i; where the decrease in value is calculated as:

$$XS_I \times g_i \times Assets_{xl}$$

g_i = A factor dependent on the credit quality of the counterparty i, as set out in appendix **TS6.1 A1**;

XS_i = The excess exposures per counterparty i as follows:

$$XS_i = max\left(\frac{E_i}{Assets_{xl}} - CT_i, 0\right);$$

E_i = Total exposure-at-default to counterparty i;

$Assets_{xl}$ = Total assets considered in the scope of the concentration risk module (including government bonds) with no allowance for loss-given-default; and

CT_i = The concentration threshold to counterparty i, which depends on the credit quality step of the counterparty i, as set out in appendix **TS6.1 A1**.

4. TS6.1 A1 – Credit Risk

Zimbabwean entities are unrated and hence the probabilities of default, spread shocks and loss given defaults proposed under ZICARP correspond to those applicable to unrated entities.

Probabilities of Default for rated entities under ZICARP

GCR Credit Rating	Credit - International Rating Scale	PD - international rating scale	PD - local rating scale
AAA (ZW)	AAA	0.01%	0.22%
AA+(ZW)	AA+	0.01%	0.22%
AA (ZW)	AA	0.02%	0.23%
AA- (ZW)	AA-	0.03%	0.25%
A+ (ZW)	A+	0.06%	0.28%
A (ZW)	A	0.09%	0.30%
A- (ZW)	A-	0.11%	0.32%
BBB+ (ZW)	BBB+	0.16%	0.37%
BBB (ZW)	BBB	0.22%	0.43%
BBB- (ZW)	BBB-	0.39%	0.60%
BB+ (ZW)	BB+	0.54%	0.75%
BB (ZW)	BB	0.81%	1.02%
BB- (ZW)	BB-	1.39%	1.60%
B+ (ZW)	B+	34%	2.75%
B (ZW)	B	5.37%	5.58%
B- (ZW)	B-	8.18%	8.38%
CCC (ZW)	CCC	26.53%	26.53%

All unrated and not subject to ZICARP	Unrated and not subject to ZICARP	10%	10%

Probabilities of Default for unrated entities that are subject to ZICARP

Solvency ratio: (Eligible Own Funds/Solvency Capital Requirement)	PD - local rating scale
>200%	0.22%
>175%	0.25%
>150%	0.30%
>125%	0.37%
>100%	1.02%
>90%	2.75%
>80%	5.58%
80% and below but above the MCR	8.38%
Below MCR	26.53%

Loss Given Default under ZICARP

Component	Standard LGD factor
Fully Cash covered	5%
Significantly over collateralised	18%
Fully collateralised	35%
Partially collateralised	43%
Unsecured	45%
Less than 50% of the company's assets are pledged as collateral for some creditors	72%
More than 50% of the company's assets are pledged as collateral for some creditors	86%
Exposures generally equity, junior debt, mezzanine debt or preference shares exposures. May also be structurally subordinated.	100%

Concentration Thresholds and g_i – Please note that there is no concentration threshold on prescribed assets.

GCR Credit Rating	Concentration Threshold (CT)	g_i
AAA (ZW)	5%	0.12
AA+(ZW)		
AA (ZW)		
AA- (ZW)		
A+ (ZW)		0.21
A (ZW)		
A- (ZW)		
BBB+ (ZW)	3%	0.27
BBB (ZW)		
BBB- (ZW)		
BB+ (ZW)		
BB (ZW)		
BB- (ZW)		
B+ (ZW)		
B (ZW)		
B- (ZW)		
CCC and all unrated	1.5%	0.73

5. TS 6.1 A2 – Symmetric adjustments to equity shocks

The equity risk capital requirement is determined by applying a capital shock (calibrated at 99.5% confidence level) to the market value of equities. However, there is need to adjust the capital shock on equities by adding to it a symmetric adjustment that prevents pro-cyclical behavior of equity exposures.

The symmetric adjustment on equity exposures is calibrated through the cycle and results in an upward adjustment on equity risk capital requirement when equities market rise and a downward adjustment on capital requirement when equities markets fall.

The purpose of the symmetric adjustment factor can therefore be summarized as follows:

a. Prevent insurance and reinsurance undertakings from being forced to raise additional capital or sell their investments due to unsustain adverse movements on the stock markets; and
b. Prevent fire sales of equities which would negatively impact equity prices and possibly destabilize the equities market when equities markets are falling.

The symmetric adjustment on equities under ZICARP is calculated as follows:

	Local Listed and Unlisted Equities
Symmetric Adjustment Factor (SAF)	$MSAF = [\frac{1}{2}(\frac{Log(CIV) - Log(AIV)}{\log(AIV)}) - 8\%]_{-8\%}^{8\%}$
Lower Bound	-8%
Upper Bound	+8%

Where:

$MSAF$ = Modified Symmetric Adjustment Factor

$Log(CIV)$ = Log of the Current All Share Index Value

AIV = Log of the Average All Share Index Value over the last 36 months

6. TS 6.4 A1: Glossary

Market risk: is the risk of loss to an insurer's assets and liabilities arising from the level or volatility of market prices of financial instruments.

Equity risk: derives from the volatility of the market prices of equities.

Property risk: arises from volatility in the market prices of property.

The counterparty default risk: under ZICARP reflects all possible losses that may arise due to unexpected default or deterioration in credit worthiness of an insurer's counterparties and debtors over the next twelve months.

Currency risk: derives from movements in the level and volatility of the currency exchange rates and their impact on the insurer's assets and liabilities.

Concentration risk: is the potential of a loss due to movements in equity, spread and default risk and property risk on assets that are considered to be not sufficiently diversified and where there is over exposure.